

Financial Engineering

Group Coursework

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Introduction

A commodity is a basic, raw material or agricultural product that can be bought, sold, and traded, characterised by its interchangeability regardless of producer. Common examples include coffee beans, crude oil, grains, and livestock.

In this research report, we focus on studying crude oil as it is one of the most traded commodities globally. Commodity derivatives (futures, options, and swaps) allow people to trade commodities without actually owning the physical product which increases accessibility to the commodity market and hence can be used for a wide variety of purposes. Therefore, We will study the features of crude oil derivatives pricing as well as their unique market structure. With a deeper understanding of the derivatives, we further analyze how we can use these instruments for both hedging and trading purposes.

Features of Commodity Derivatives Pricing

In the classical Hotelling (1931) framework, the price of an exhaustible resource is expected to grow at the risk-free rate, reflecting intertemporal scarcity. Under this benchmark, resource prices follow a deterministic upward path.

However, empirical evidence from crude oil markets reveals substantial deviations from this prediction. Oil prices exhibit pronounced volatility, frequent backwardation, and time-varying risk premia. These deviations arise from short-run inventory dynamics, storage constraints, and stochastic convenience yields, which introduce state-dependent pricing effects absent from the static Hotelling model.

1.1 Inventory and Storage Cost

Inventory plays a central quantitative role in crude oil pricing by linking physical market conditions to derivative valuation through the cost-of-carry relation. Under no-arbitrage conditions, the futures price $F_t(T)$ with maturity T satisfies:

$$F_t(T) = S_t \exp [(r_t + u_t - y_t)(T - t)] ,$$

where S_t denotes the spot price, r_t the risk-free rate, u_t storage costs, and y_t the convenience yield. Inventory affects pricing primarily through its influence on the convenience yield, which reflects the marginal value of holding physical oil.

From a theoretical perspective, crude oil inventories function as a buffer stock that mitigates short-term

supply-demand imbalances. Let I_t denote aggregate inventory at time t . Empirically, the convenience yield is decreasing in inventory:

$$\frac{\partial y_t}{\partial I_t} < 0.$$

Inventory also affects price dynamics through buffer-stock effects. When I_t is high, temporary supply are absorbed with relatively small spot price adjustments. When I_t is low, similar shocks generate larger price responses, implying volatility. This mechanism provides a natural explanation for periods of elevated oil price volatility during tight market conditions.

Storage costs u_t represent the explicit and implicit costs of holding physical oil, including warehousing, insurance, maintenance, and financing expenses. Unlike financial assets, crude oil requires physical capacity for storage, which is limited and often location-specific. As a result, storage costs may vary across time and across delivery points.

In the cost-of-carry framework, storage costs enter additively into the carry component $(r_t + u_t - y_t)$. A higher u_t increases the required futures price relative to spot, all else equal. Formally,

$$\frac{\partial F_t(T)}{\partial u_t} > 0.$$

When storage capacity becomes constrained, marginal storage costs can rise sharply, introducing non-linearities into the pricing relation.

1.2 Stochastic Convenience Yield

The convenience yield, y_t , captures the implicit benefit of holding physical oil. Economically, it reflects the value of immediate availability, operational flexibility, and protection against supply disruptions.

Within the cost-of-carry framework in above section, the convenience yield enters as a negative component of net carry:

$$\frac{\partial F_t(T)}{\partial y_t} < 0.$$

From a quantitative perspective, the convenience yield is treated as a stochastic state variable. In affine term-structure models of commodity futures, it is commonly specified as a mean-reverting diffusion under the risk-neutral measure. In particular, Casassus and Collin-Dufresne (2005) show that the convenience yield can be represented as

$$dy_t = \kappa_y^Q (\theta_y^Q - y_t)dt + \sigma_y dW_t^Q,$$

where κ_y^Q denotes the speed of mean reversion, θ_y^Q the long-run mean, and σ_y the volatility of scarcity shocks.

Within their affine framework, the convenience yield interacts with the short rate and the spot price process, and is embedded as a latent factor in the futures term structure. Their results imply that variations in the slope and curvature of the crude oil futures curve can be interpreted as reflecting changes in the underlying stochastic convenience yield process. Convenience yield serves as a state variable that transfers scarcity to financial prices. The stochastic nature of y_t therefore introduces time variation in carry components and expected returns.

1.3 Term Structure: Contango and Backwardation

The sign of the net carry component $(r_t + u_t - y_t)$ expression reflects underlying economic conditions rather than purely mechanical cost-of-carry considerations.

Contango occurs when inventories are abundant and the marginal value of physical availability is low. In such states, the convenience yield y_t is small relative to financing and storage costs. Formally, when

$$y_t < r_t + u_t,$$

the term structure is upward sloping. These conditions typically prevail during periods of weak demand, high production, or when storage capacity is underutilized. High inventory levels compress scarcity premia and reduce the benefit of holding spot oil.

Backwardation emerges when the convenience yield exceeds carrying costs:

$$y_t > r_t + u_t.$$

This configuration reflects tight market conditions. When inventories approach low levels, the marginal value of immediate availability rises sharply. Refiners and industrial users are willing to pay a premium for prompt delivery, pushing spot prices above futures prices.

Because the convenience yield is stochastic and state-dependent, the oil futures curve can switch between contango and backwardation as inventory conditions changes.

Crude oil markets exhibit backwardation more frequently than many storable commodities due to several structural characteristics. First, storage capacity for crude oil is limited and location-specific. As inventories decline, marginal storage availability becomes increasingly constrained, leading to increases in the scarcity value in the convenience yield. Second, short run oil demand is relatively inelastic. Consequently, supply disruptions generate disproportionately large spot price responses, increasing the value of immediate physical availability.

An alternative explanation is provided by Litzenberger and Rabinowitz (1995), who argue that persistent backwardation in crude oil cannot be fully explained by storage-based convenience-yield models alone. They emphasize that crude oil production involves intertemporal extraction decisions, and that underground reserves effectively embed a real option. In this framework, backwardation arises not only from inventory scarcity, but also from optimal production timing and the option value embedded in oil reserves.

In summary, inventory dynamics, storage constraints, stochastic convenience yield, and term structure effects jointly determine the pricing of crude oil futures. These mechanisms are not instrument-specific because most oil derivatives such as options, swaps, and forwards, are linked to the futures curve, variations in these structural components propagate across derivative instruments. The analysis of futures pricing therefore provides the foundation for understanding the valuation and risk premia of other oil derivatives.

Hedging Strategies from Features of Crude Oil Derivatives Pricing

Hedging strategies are essential tools in the crude oil market to mitigate the risks of price fluctuations. By using instruments like futures, swaps, options, and spread hedging, producers and consumers can secure stable revenues. These strategies allow them to manage volatility, safeguard against market uncertainties, and provide better financial planning for the future.

2.1 Futures

Crude oil futures are financial instruments used to manage price risk by allowing participants to lock in prices for oil to be delivered in the future. These futures contracts help oil producers and consumers hedge against price volatility by securing a future price. The cost-of-carry model explains how crude oil futures are priced:

$$F_t(T) = S_t \exp [(r_t + u_t - y_t)(T - t)]$$

In this equation, $F_t(T)$ is the futures price at time t for a contract expiring at T , S_t is the spot price of

crude oil, r_t is the risk-free interest rate, u_t represents the costs of storage and insurance, and y_t is the convenience yield, which reflects the benefit of holding physical oil rather than the futures contract. The convenience yield fluctuates based on market factors such as supply and demand, inventory levels, and geopolitical events. It often increases during periods of tight supply, causing futures prices to rise relative to spot prices (Alquist et al., 2014). As shown in Figure 1, the chart illustrates the price movements of crude oil spot prices and futures prices with varying maturities (1-month, 3-month, and 12-month futures) over time, highlighting the impact of convenience yield and market conditions on the futures pricing structure.

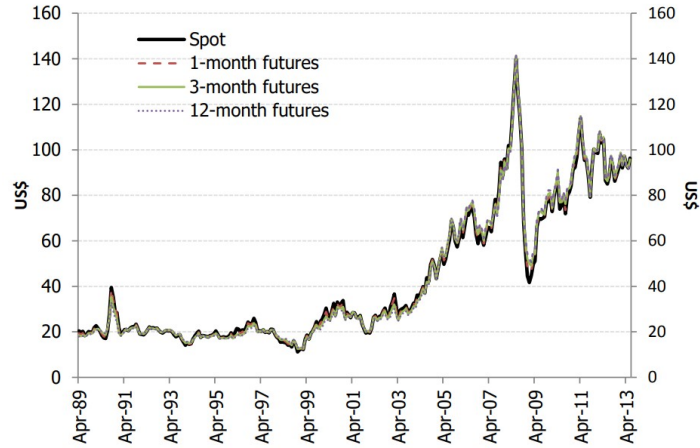


Figure 1: The chart shows the price movements of crude oil spot prices and futures prices with varying maturities (1-month, 3-month, and 12-month futures) over time.

In practice, crude oil futures are mainly used for hedging. Oil producers, like drillers, face price uncertainty and use futures contracts to lock in prices for future sales. By selling futures contracts, producers ensure a consistent price for their oil, stabilizing future revenue despite potential price declines. On the other hand, businesses that need to purchase oil, such as refineries, use futures contracts to lock in future purchase prices, protecting themselves from rising costs.

The success of hedging with futures depends on several factors. One of the most important is the hedge ratio, which determines the proportion of futures contracts needed to offset price changes in the underlying asset. The optimal hedge ratio minimizes risk and maximizes the hedge's effectiveness, calculated using the historical correlation between spot and futures prices. The formula for the hedge ratio is:

$$h^* = \frac{\text{Cov}(S_t, F_t)}{\text{Var}(F_t)},$$

where $\text{Cov}(S_t, F_t)$ is the covariance between spot and futures prices, and $\text{Var}(F_t)$ is the variance of futures prices. Another key factor is basis risk, which arises when the spot and futures prices do not move perfectly in sync.

Futures contracts are a valuable tool for managing price risk, but their success depends on proper implementation, including calculating the hedge ratio accurately and aligning the timing of the hedge with the exposure period.

2.2 Swaps

An oil swap is a bilateral agreement where one party pays a fixed price while the other pays a floating price based on the market's forward curve. This mechanism is widely used to hedge against price volatility. The fixed price helps stabilize cash flows, while the floating price adjusts to market movements.

The figure below illustrates how an oil swap works to hedge price risk. The red dashed line represents the unhedged spot price, which directly fluctuates with the market. If the spot price decreases, revenue falls; if it increases, revenue rises. In contrast, the blue solid line represents the net effective price after entering a fixed-price swap. Regardless of the spot price movement, the net effective price remains anchored at the fixed swap level.

If the spot price falls below the fixed swap price, the swap generates a positive payoff to offset the loss from the lower spot price. Conversely, if the spot price rises above the fixed swap price, the swap incurs a loss, but the overall price remains close to the swap target. This makes swaps a useful tool for hedging, providing stability for oil producers and consumers alike. Producers lock in stable revenues, while consumers hedge against rising prices.

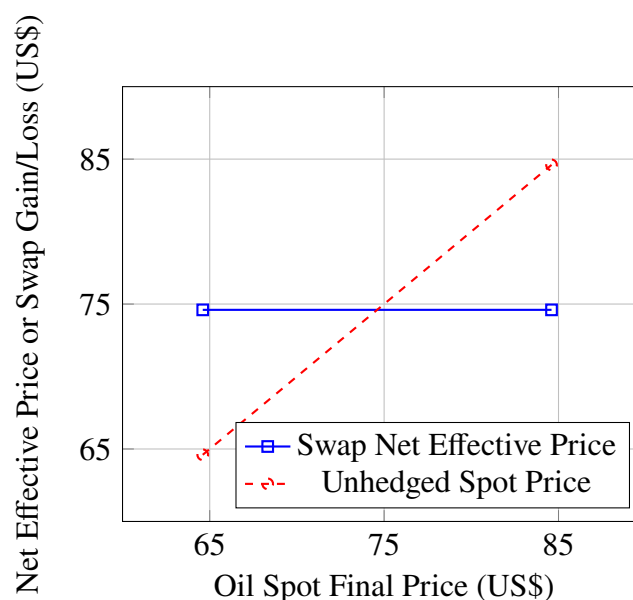


Figure 2: Swap hedging results under different spot price scenarios, comparing net effective prices with unhedged spot prices.

2.3 Options

The relatively high premium cost of plain vanilla options is the reason why many crude oil producers and consumers adopt structured option strategies. Among all strategies, collars are particularly prevalent.

A collar strategy means purchasing a put option to establish a price floor while simultaneously selling a call option to finance the premium cost, which is often called a “zero-cost collar”. For oil producers, this structure ensures downside protection against falling prices, while the written call caps potential gains if prices rise above the call strike. The result is revenue fluctuating within a price range.

Compared with an alone put option, the collar reduces hedging costs but sacrifices unlimited upside potential. For producers who prioritise revenue stability over speculative gains, this trade-off is often acceptable. Indeed, publicly listed oil and gas companies frequently disclose the use of collars in their hedging programs to stabilise cash flows and support capital budgeting decisions. However, collars do not eliminate other sources of uncertainty. Basis risk remains if the options are written on benchmark crude prices that differ from the firm’s physical exposure. Additionally, extreme price movements may still lead to opportunity costs when upside gains are capped.

Therefore, collars represent a cost-efficient and widely adopted hedging mechanism in crude oil markets, balancing downside protection with premium affordability while managing price risk. To illustrate the economic effect of a collar more clearly, Figure 3 shows how downside protection at K_{put} and an upside cap at K_{call} convert linear exposure into a bounded revenue range relative to the unhedged position.

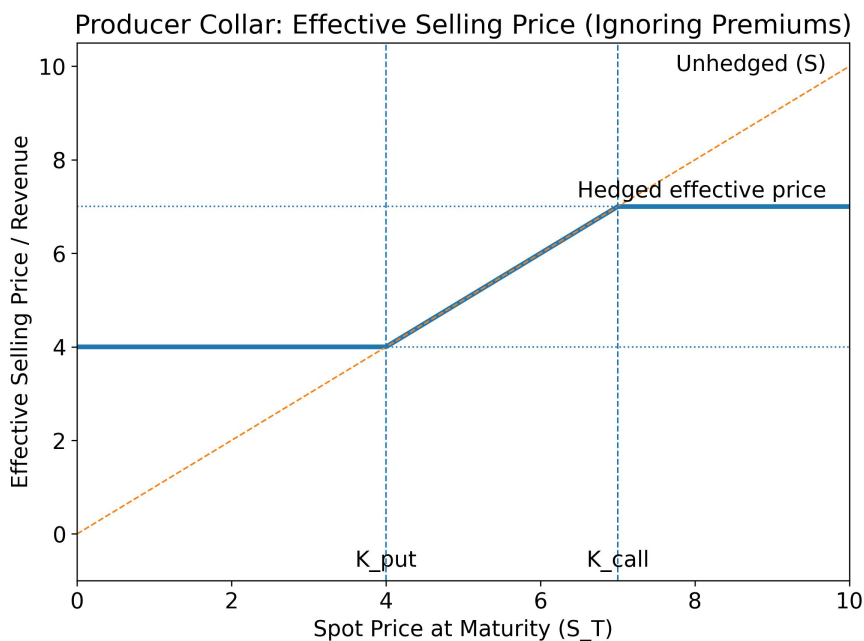


Figure 3: Producer collar strategy

2.4 Spread hedging

The key insight is that refinery profitability depends on the margin between output and input prices. Even if crude oil prices decline, profits may also fall if refined product prices decline by a larger amount. Conversely, margins may increase when product demand increases more compared to crude supply. Therefore, refiners are exposed to volatility in the combination of crack spread rather than to crude oil prices alone.

To hedge this exposure, refiners may construct positions in multiple derivative instruments rather than relying on a single contract type. In the case of a 3:2:1 crack spread hedge, a refiner can take a long position in crude oil futures and short positions in gasoline and heating oil futures in appropriate proportions. This structure locks in the margin implied by current futures prices and reduces the sensitivity of profits to changes in relative prices. Also, swaps and options can be used as tools to deal with crack spread exposure. In some cases, firms combine futures for baseline hedging with options to manage tail risk, thereby constructing layered risk management strategies.

Overall, spread hedging reflects a fundamental feature of commodity markets. Risk frequently arises from relative price movements rather than from outright price levels. By targeting price differentials between inputs and outputs, regions, or maturities, firms can stabilise operating margins and reduce earnings volatility.

Trading Strategies from Features of Crude Oil Derivatives Pricing

Traders, compared to hedgers, have a different mentality as they look to profit off oil prices by using various derivatives to express their views on the oil market. Traders utilise a wide array of data - macroeconomic data, news events, supply and demand data, etc - to form views around the market and trade on those ideas.

3.1 Futures Spread (Roll Yield) Trading

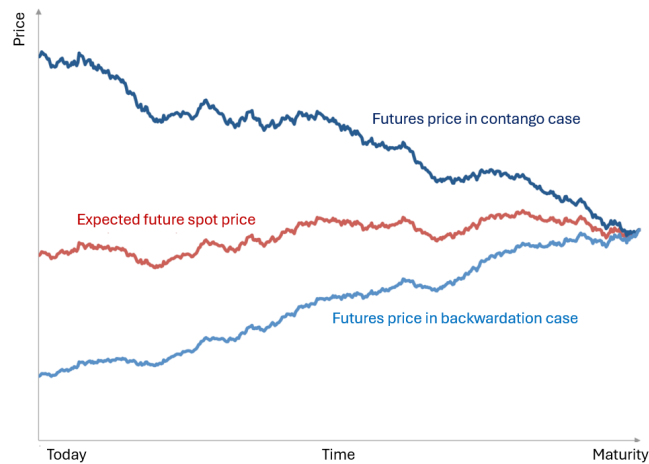


Figure 4: Futures prices of crude oil futures contracts in contango versus backwardation situations

Roll yield is the return of an investor by adjusting their futures position by moving from a near-term contract to a longer-term contract. The roll yield can be calculated as below:

$$\text{Roll Yield} = (\text{Futures Price}_t - \text{Futures Price}_{t-1}) - (\text{Spot Price}_t - \text{Spot Price}_{t-1})$$

When traders buy futures for speculation, they must buy the underlying asset (crude oil) at a set future date unless they sell the position before the delivery date. Many traders do not want to take delivery of the physical asset represented by the futures contract since they are purely price speculators or do not have the capacity to build warehouses to store crude oil. As such, they can either close their position before expiry or roll (renew) their near-term expiring futures contracts into futures contracts with expiration dates further in the future. To roll a contract, traders must first settle the gain or loss of the original contract.

1 Backwardation Scenario

As mentioned in section 1.3, during a backwardation, the future price of crude oil is below the spot price (Figure 4). As such, a trader can profit if they roll their current contract to a contract with a later expiration date since the trader effectively pays a lesser price as compared to the spot price for the underlying asset represented by the new futures contract.

2 Contango Scenario

When the market is in contango, the future price of crude oil is above the spot price (Figure 4); longer-dated futures are more expensive than near-dated ones. In this case, if a trader rolls over their contract, they effectively make a loss. To make a profit, traders can enter a short position by selling a futures contract. Normally, traders would short the front-month futures contract compared to a longer-dated contract, as it has the nearest expiration date, and thus can converge to the future spot quickly which minimises duration exposure risk.

3 Calendar Spread Trading

Traders can also trade on the shape of the futures curve (Figure 4) instead of the price of oil to reduce directional oil risk. This can be performed via a calendar spread trade strategy where the trader buys a crude oil futures contract while selling another crude oil futures contract with a different expiration date simultaneously. Profits arise because the short-term option loses value faster than the long-term option, widening the spread, for example, in the case of theta harvesting.

3.2 Location and Storage Arbitrage Trading

Location arbitrage trading in crude oil is defined as profiting from price discrepancies for oil of similar grades across different geographical markets. In this strategy, traders profit by simultaneously buying oil at a lower-priced region and selling it in a higher-priced region. Arbitrage opportunities exist because oil is a physical commodity and has to travel vast distances to reach different regions, which creates inefficiencies.

1 Physical Arbitrage (Location)

To execute this strategy, a trader can physically buy West Texas Intermediate (WTI) - US oil market benchmark - and export it to Europe and sell it at the Brent Crude price, which is the benchmark for the light oil market in Europe, Africa, and the Middle East. The success of this strategy hinges on the landed value, which is the total cost of transporting oil to its destination. Traders have to take into account freight and logistics, time structure, benchmark spreads, and other overhead costs to identify if the price spread between different markets exceeds logistical and transportation costs.

2 Statistical Arbitrage (Location)

Instead of physically buying crude oil, traders can also execute an arbitrage via futures contracts. A very common way is to simultaneously take opposite positions in 2 geographically different benchmarks. For example, if the price of a benchmark (WTI) is higher than another (Brent) compared to historical averages, a trader trades the larger than normal spread by taking a long position on the undervalued futures contract (WTI for example) and short the overvalued futures contract (Brent for example) and make a profit if the spread reverts to its historical mean.

3 Storage Arbitrage

A storage arbitrage trading strategy (or a time arbitrage) involves buying the physical oil at a low spot price, storing the oil, and then simultaneously selling a futures contract to sell an equivalent amount of oil at a higher price with an expiration date later in the future. This strategy works in a market structure experiencing contango, as futures prices are higher than the immediate spot price. However, traders have to calculate the "cost of carry", which includes storage fees, financing, and insurance of the physical oil. The profit earned by the trader is the difference between the sale price of the oil at delivery and the purchase price and cost of carry of the same oil.

3.3 Crack Spread Trading

Seasonal patterns play a huge role in speculators' decision making. Figure 5 is a chart of crude oil price changes showing distinct spikes in January and June-July, driven by winter heating oil demand and summer gasoline demand for travel.



Figure 5: Crude Oil Price (WTI)

The seasonal demand is more pronounced in the refined products. In summer, gasoline demand surges; in winter, heating oil demand spikes. Crude oil prices may also increase due to refinery demand but not as much as heating oil and gasoline, which widens the crack spread. Figure 6 below supports the

idea that the 3:2:1 crack spread widens during summer (gasoline-led) and winter (heating oil-led), creating opportunities for bull spread trades.

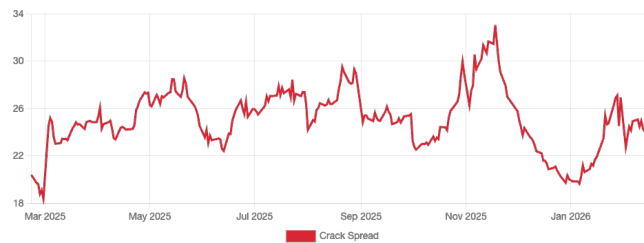


Figure 6: 3:2:1 Crack Spread

Investors can capture this by executing a 3:2:1 ratio bull crack spread as mentioned in Appendix () in late spring (for the summer driving season) or early autumn (for winter heating demand) to see a profit in the spread as prices of gasoline and heating oil spike. Profit is realised if:

1. Gasoline / heating oil rises faster than crude, or
2. Gasoline / heating oil rises while crude stays flat, or
3. Crude falls less than gasoline / heating oil rises

3.4 Straddle/Volatility Trading

Crude oil is a highly volatile commodity, and its price can be significantly affected by events that have the potential to disrupt supply, including geopolitical and weather-related developments. Its demand and supply are inelastic: it takes years to develop new oil production capacity, and as a key economic necessity, consumers cannot immediately find substitutes when prices rise. As a result, instead of quantity adjusting, price must jump sharply to clear the market.

Historically, political events have caused significant oil price spikes and elevated volatility. A clear example is the direct Iran-US conflict in June 2025. Following Israel's airstrikes on Iranian facilities, Brent crude prices surged more than 10%, reaching a six-month high of \$74/bbl. Notably, there was no immediate physical disruption to oil flows but created fear about future supply interruption.

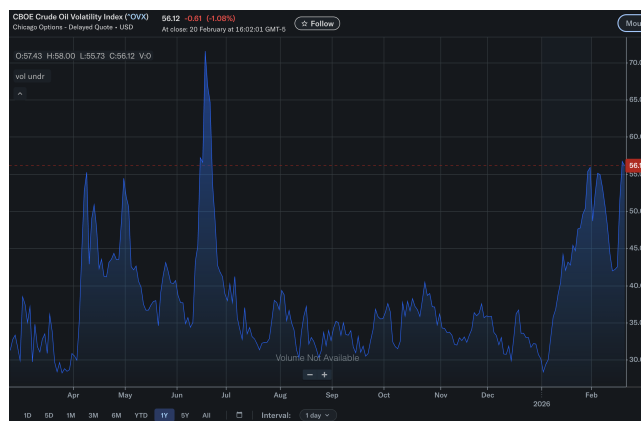


Figure 7: CBOE Crude Oil Volatility Index

Traders anticipating such volatility can profit using options like straddles or strangles, depending on their timing and market conditions.

A straddle involves buying an at-the-money (ATM) call and an ATM put with the same strike price and expiration. It profits from large price movements in either direction and is ideal when a moderate but likely price move is expected.

A strangle involves buying out-of-the-money (OTM) call and put options with different strikes. It has a lower premium than a straddle but requires a larger price move to profit. It is often used by late buyers chasing extreme price swings or panic-driven moves.

In the June 2020 Iran example, speculators anticipating the price spike before headlines broke would benefit from a straddle, while traders entering after the initial surge could use a strangle to profit from further extreme movements without paying the higher straddle premium.

Conclusion

The instruments and strategies analysed in this report, and their combinations derive their utility and profitability from the fundamental characteristics of oil as a geographically dispersed and economically essential commodity. The cost-of-carry framework demonstrates how inventory dynamics, storage costs, and stochastic convenience yields jointly determine the term structure of futures prices, explaining the frequent regime shifts between contango and backwardation that distinguish crude oil from financial assets.

For hedgers, these pricing features directly inform risk management practices. Producers and consumers utilise futures for price locking, swaps for cash flow stability, and option-based strategies such as collars for balanced downside protection. Critically, spread hedging recognises that for refiners and intermediaries, relative price movements—crack spreads, location differentials, and calendar spreads—often matter more than absolute price levels. For traders, the same structural features create

profit opportunities through roll yield trading, location and storage arbitrage, seasonal crack spread strategies, and volatility positioning around geopolitical events.

A thorough understanding of these pricing features is therefore essential not only for effective risk management but also for the identification and execution of viable trading strategies in one of the world's most liquid and strategically important commodity markets.

Appendix

1.1 Further analysis of spread and linkage to hedging and trading

In real-world crude oil markets, not all participants are primarily exposed to direct price risk. For many firms, particularly refiners and intermediaries, the main source of uncertainty arises from fluctuations in relative prices rather than from the absolute level of oil prices. In such cases, hedging strategies are designed to stabilise price differentials. This approach is commonly referred to as spread hedging. An example in the oil industry is the crack spread, which represents the refining margin. Refiners purchase crude oil as an input and sell refined products such as gasoline and diesel. A simplified representation of refinery profit can be shown by the following equation:

$$\Pi = 2P_{\text{gasoline}} + P_{\text{diesel}} - 3P_{\text{crude}} - C, \quad (1)$$

where P_{gasoline} , P_{diesel} , and P_{crude} denote market prices and C represents processing costs. The coefficients reflect a stylised 3:2:1 crack spread, meaning that three barrels of crude oil produce approximately two barrels of gasoline and one barrel of diesel.

Besides the above case, location spreads, such as the important Brent-WTI differential, are also relevant for firms exposed to regional pricing differences. Since Brent and WTI benchmarks are influenced by transportation constraints, storage capacity and regional supply and demand conditions, Firms need to hedge this exposure that can materially affect revenues and costs.

In addition, calendar spreads represent another important dimension of spread risk. Changes in the term structure of oil futures, whether in contango or backwardation, affect inventory valuation and rolling hedge costs. Market participants can hedge the risk of adverse shifts in the futures curve by taking positions in contracts with different maturities. Such strategies are particularly relevant in crude oil markets, where storage costs and convenience yields influence the shape of the futures curve.

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